

Problem Statement No.8- Nutrition Agent

The Challenge - As health awareness surges, individuals increasingly demand tailored nutritional guidance, yet current tools rely on generic diet plans, lack real-time adaptability, and ignore holistic lifestyle factors like cultural preferences, allergies, and dynamic health conditions. Furthermore, professional nutritionists face severe scaling limitations due to time and resource constraints. Generative AI unlocks a breakthrough opportunity to solve this by powering an intelligent, interactive virtual assistant. By integrating natural language processing, vector retrieval, and enterprise foundation models, an advanced assistant can generate dynamic meal plans, recommend smart food swaps, and explain nutritional choices grounded in clinical data. This project develops "NutriBot," a smart AI nutrition assistant using custom Flask workflows and agentic orchestration that:

- **Understands** user inputs through structured profiling and multi-agent interaction handlers.
- **Generates** hyper-personalized meal plans and dynamic macronutrient targets tailored to individual goals and health restrictions.
- **Offers** contextual explanations and dietary insights backed by local vector knowledge retrieval.
- **Adapts** suggestions continuously through persistent food logging and rolling session feedback loops.

Technology - Built using Python, Flask, FAISS vector search, and mandated integration with **IBM watsonx.ai** leveraging the **ibm/granite-3-8b-instruct** foundation model for enterprise-grade reliability and governance.